EXPLORATORY DATA ANALYSIS

Chapter 25

Communicating Insights and Analytical Uncertainty

Turning evidence into clear, responsible, useful communication.

A practical guide for analytical reports, presentations, and client conversations.

Good analysis is not complete when the calculations are correct. It is complete when people can understand what the evidence means, what it does not mean, and what they can reasonably do next.

1. Learning Objectives

- Prioritize findings according to relevance, evidence strength, and usefulness.
- Communicate results in clear language for technical and non-technical audiences.
- Distinguish observations, interpretations, hypotheses, and causal claims.
- Explain uncertainty, variability, limitations, and sensitivity without false confidence.
- Choose appropriate language for estimates, comparisons, associations, and incomplete evidence.
- Answer questions and challenges without overstating or becoming defensive.
- Build a concise presentation or verbal walkthrough of an EDA project.
- Translate analysis into evidence-linked next steps while preserving analytical integrity.

2. Review of Earlier Chapters

This review revisits the ideas that support responsible communication. Chapter 24, the immediately previous chapter, is reviewed separately in Section 3.

- Chapters 1–3: EDA is investigation, not automatic explanation. Define the observational unit, understand metadata, and assess quality before interpreting results.
- Chapters 4–9: numerical and categorical summaries, time variables, distributions, quantiles, skewness, and kurtosis describe different aspects of data. Choose statistics that fit the distribution.
- Chapters 10–12: charts are analytical tools. Scales, bins, encodings, facets, and aggregation can change the impression a reader receives.
- Chapters 13–15: correlation, association, covariance, dependence, and grouped analysis are not interchangeable. Group definitions and denominators matter.
- Chapters 16–17: an unusual observation is a signal to investigate, not proof of an error. Document conclusions and retain context.
- Chapter 18: high-dimensional exploration creates many possible patterns; selection and multiple comparisons make overinterpretation easy.
- Chapter 19: temporal patterns depend on timezone, cadence, coverage, and aggregation. A trend may also reflect collection changes.
- Chapter 20: sampling, missingness, nonresponse, and selection bias constrain generalization.
- Chapter 21: automated profiles and alerts identify things to inspect; they do not establish meaning or cause.
- Chapter 22: domain context determines what rows and variables represent, which comparisons are meaningful, and which privacy constraints apply.
- Chapter 23: reproducibility makes data, code, settings, and analytical decisions traceable.

3. Review of Chapter 24 — Building a Professional EDA Report

Chapter 24 organized analysis into a reader-focused report: executive summary, objectives, dataset scope, data quality, methodology, findings, limitations, recommendations, and appendix. It emphasized clear tables and figures, captions, meaningful denominators, and the distinction between observation, interpretation, and possible explanation.

Expected reasoning for the previous assignment/project: a strong client-ready report identifies its audience and purpose, explains what one row represents, states the analytical population and filters, quantifies data-quality issues, supports findings with reproducible tables and figures, and discloses limitations. Every recommendation should connect to evidence or identify a gap requiring further investigation.

A report can be technically complete yet difficult to use if its central message is buried. This chapter focuses on communicating that message clearly and honestly.

4. What Is an Insight?

A result is a calculated value or observed pattern. An insight is a result interpreted in context so that it helps someone understand a situation or consider a decision.

Result: The median delivery time is 3.2 days.

More informative insight: The median delivery time is 3.2 days, but the 90th percentile is 8.1 days. Most deliveries are relatively quick, while a smaller group experiences substantially longer waits.

The second statement helps the reader see why the median alone may not describe the full customer experience.

4.1 Components of a useful insight

- Evidence: a statistic, comparison, chart, or quality finding.
- Context: the population, period, unit, and definition.
- Meaning: why the pattern matters for the question.
- Boundary: what the evidence cannot establish.
- Implication: a reasonable next question or action.

Not every interesting pattern is important. A tiny difference may be statistically visible but irrelevant. A large-looking difference may be unstable or driven by a small sample.

5. Prioritizing Findings

An EDA project can generate dozens of observations. A reader usually cannot act on all of them at once. Select findings rather than presenting every output.

5.1 A practical prioritization framework

- Relevance: does the finding answer the original question or matter to the audience?
- Evidence strength: is it supported by adequate data and more than one reasonable check?
- Magnitude: is the difference large enough to matter in context?
- Reliability: does it persist under reasonable alternative definitions or filters?
- Actionability: can the reader investigate or act on it?
- Risk: could misunderstanding the finding lead to a harmful decision?

A report often has three to five main findings, with secondary details in supporting sections or an appendix. This is a communication guideline, not a statistical rule.

5.2 Do not prioritize by visual drama

A striking outlier or colorful heatmap can attract attention even when it is not important. Prioritize according to the question and evidence, not which chart looks most impressive.

5.3 Keep a finding register

```
finding_register = [
    {
        "finding": "Delivery times have a long upper tail",
        "evidence": "Median 3.2 days; 90th percentile 8.1 days",
        "relevance": "Customer experience and service reliability",
        "caveat": "Only completed deliveries are included",
        "next_step": "Compare delays by region and carrier"
    }
]
```

6. Communicating Numbers in Plain Language

Readers need to understand what a number represents before they can use it.

6.1 Include the essentials

- Name the measure and its unit.
- Identify the group or population being described.

- Give the sample size or denominator when relevant.
- Provide a comparison or reference point when useful.
- Explain the practical meaning without adding an unsupported cause.

Weak: 'The rate is 12%.'

Clearer: 'Among 2,400 submitted applications, 288 were incomplete, giving an incompleteness rate of 12%. This describes submitted applications in this dataset; it does not necessarily represent all applicants.'

6.2 Absolute and relative differences

A relative increase can sound large when the starting value is small. If a rate rises from 2% to 3%, that is an increase of 1 percentage point, or a 50% relative increase. These statements describe the same change but create different impressions.

6.3 Percentages need denominators

A group with 30 cases out of 100 has a 30% rate. A group with 300 cases out of 2,000 has a 15% rate. Report numerator and denominator when group size affects interpretation.

6.4 Use sensible precision

- Avoid long decimal strings that imply more precision than the data supports.
- Use consistent rounding across comparable groups.
- Do not round in a way that changes the conclusion.
- Keep units visible: days, BDT, kilograms, records, percentage points.
- Do not mix percentage change with percentage-point change.

7. Observation, Interpretation, Hypothesis, and Causation

These levels of claim have different evidential requirements.

- **Observation:** a pattern directly described by the data.
- **Interpretation:** what the pattern means in the known context.
- **Hypothesis:** a possible explanation that needs further checking.
- **Causal claim:** a statement that changing one factor produces a change in another; this requires an appropriate design and assumptions.

Observation: Customers who used the mobile app placed more orders per month than customers who did not.

Interpretation: App users in this dataset are a higher-activity group.

Hypothesis: App features may encourage repeat purchasing, or more active customers may be more likely to install the app.

Causal claim: Installing the app increases monthly orders. The observed comparison alone does not establish this.

The direction could be reversed, or both app use and order frequency could relate to customer tenure, marketing exposure, income, or another factor.

7.1 Correlation is not causation

An association tells us that variables vary together in the observed data. It does not show by itself that one causes the other. Confounding, reverse direction, selection effects, measurement choices, and chance patterns may contribute.

7.2 Use language that matches the evidence

- Prefer 'was associated with' over 'caused' for observational comparisons.
- Use 'in this dataset' when scope is limited to the observed records.
- Use 'is consistent with' when evidence supports a possibility but does not settle it.

- Use 'may reflect' for a plausible explanation not yet confirmed.
- Say 'we cannot determine from these data' when required evidence is absent.

8. Understanding Analytical Uncertainty

Uncertainty is not a weakness to hide. It tells the reader how confidently a result can be interpreted and how far it can be generalized.

8.1 Different sources of uncertainty

- Sampling uncertainty: the observed sample may differ from the target population.
- Measurement uncertainty: values may be imprecise, misclassified, or based on proxies.
- Missing-data uncertainty: missingness may be systematic and change the observed pattern.
- Selection uncertainty: included records may differ from excluded or unreachable units.
- Method uncertainty: conclusions may depend on statistical choices or assumptions.
- Temporal uncertainty: patterns may change across periods or after a process change.
- Definition uncertainty: reasonable metric definitions may yield different results.

8.2 Variability is not the same as uncertainty

Variability describes how much individual observations differ. Uncertainty describes how much we do not know about a quantity or conclusion. A dataset can have highly variable delivery times while its median is estimated precisely; a low-variability sample can still be unrepresentative.

8.3 Descriptive EDA and inferential uncertainty

EDA often describes the records in hand. A calculated mean for a complete dataset is a descriptive result for those records, subject to measurement and quality limitations. It is not automatically an estimate of a wider population.

Confidence intervals, hypothesis tests, and margins of error require a suitable inferential framework. They are not automatic decorations for every EDA chart. Their interpretation depends on sampling design, assumptions, and target quantity.

8.4 Avoid false precision

A result such as 43.728491% may look exact even when data are incomplete or the sample is biased. Decimal places do not remove uncertainty about measurement, selection, or representativeness.

9. Sensitivity and Robustness

A conclusion is more reassuring when it remains broadly similar under reasonable alternative choices. Sensitivity analysis checks whether the message depends heavily on one analytical decision.

9.1 Common sensitivity checks

- Compare mean and median when extreme values may influence the mean.
- Compare results with and without a clearly defined subset, explaining why both versions matter.
- Use alternative but defensible date windows.
- Compare complete-case results with a suitable missing-data-aware description.
- Check whether group patterns remain when sizes and denominators are considered.
- Compare reasonable histogram bin widths or use an ECDF.
- Check whether one extreme observation dominates a relationship.

9.2 Example: a change in average value

Suppose average transaction amount rises after a small number of very large transactions are removed. This does not show automatically that remaining transactions became more valuable. The removal changed the population being summarized. Report original and filtered results, the exclusion rule, number affected, and the question each summary answers.

9.3 Robustness does not prove truth

A result that survives several checks is less dependent on those choices. It can still be affected by poor measurement, selection bias, missing variables, or a flawed definition.

10. Communicating Group Comparisons

- State how groups were defined and whether membership can overlap.
- Show group sizes and relevant denominators.
- Compare rates or distributions when raw counts are dominated by group size.
- Check whether groups differ in tenure, region, exposure, or other relevant composition.
- Avoid ranking tiny groups as if their estimates were equally stable.
- Do not treat an aggregate comparison as proof of an individual-level pattern.

Simpson's paradox illustrates how an aggregate relationship can differ from relationships within subgroups. When a result changes after grouping, explain the composition difference rather than choosing whichever result supports a preferred story.

11. Communicating Time-Based Findings

- State the exact period and timezone.
- Explain whether the chart shows events, unique entities, totals, averages, or rates.
- Distinguish missing periods from periods with zero events.
- Mention changes in tracking, instrumentation, definitions, or coverage.
- Do not call a short-term fluctuation a trend without enough context.
- Distinguish recurring seasonality from one-time events and structural changes.

Example: 'Recorded support tickets fell by 20% in July' is incomplete if the ticketing system was unavailable for several days. The interruption may explain the lower recorded count.

12. Communicating Data Quality and Missingness

Data quality findings can be central insights, not merely technical footnotes.

12.1 Explain why a quality issue matters

Instead of saying '18% of income values are missing', explain which analysis is affected, whether missingness differs across groups, and whether the mechanism is known or unknown.

12.2 Do not invent the meaning of missing values

A blank may mean unknown, not collected, not applicable, intentionally withheld, or lost during processing. Unless documentation confirms the meaning, do not choose one explanation without evidence.

12.3 State the limits of observed data

If the dataset includes only completed deliveries, the report cannot describe failed or abandoned deliveries unless those records exist elsewhere. A survey of respondents does not automatically represent nonrespondents.

13. Choosing the Right Communication Format

- Executive summary: key findings, risks, and next steps for a time-limited decision-maker.
- Full report: structured evidence, methodology, limitations, and supporting detail.
- Presentation: a short narrative with a small number of high-value visuals.
- Dashboard: recurring monitoring of defined metrics; it needs clear definitions and governance.
- Technical appendix: formulas, detailed tables, validation checks, and reproducibility notes.
- Client conversation: explain the main result, ask what decision it informs, and clarify assumptions.

13.1 One message per visual

A slide or chart should have one main message. Use a headline that states the takeaway, then show the visual evidence. Avoid filling a slide with unrelated plots.

13.2 A short presentation structure

```
1. The question and why it matters
2. What data was examined
3. Important data-quality caveats
4. Three main findings
5. What the findings may mean
6. What remains uncertain
7. Recommended next checks or actions
```

14. Answering Questions and Handling Challenges

A client may ask why a number changed, why a group was excluded, or whether a finding proves a theory. A professional response is clear, calm, and evidence-led.

14.1 A four-step response

- Clarify the question if ambiguous.
- State what the analysis directly shows.
- Explain the relevant method, assumption, or limitation.
- Offer a concrete check or additional data source if the question cannot yet be answered.

Question: 'Does this prove the campaign worked?'

Responsible response: 'Orders increased during the campaign period, but this analysis does not isolate the campaign's effect. Seasonality, pricing, stock availability, and other changes could contribute. We would need a stronger comparison design or additional data to estimate the campaign's impact.'

14.2 When you do not know

It is acceptable—and often professionally important—to say that you do not know. Be specific about what is missing and how it could be investigated.

Useful wording: 'I cannot determine that from the current dataset. We would need [specific field or comparison] to test it.'

14.3 Avoid defensive communication

- Do not hide a mistake; correct it and explain its impact.
- Do not use technical language to silence questions.
- Do not promise certainty that the data cannot provide.
- Do not change a definition silently to make a result look better.
- Treat a challenge as an opportunity to check assumptions.

15. Ethical and Responsible Communication

- Protect personal, confidential, and sensitive information.
- Avoid identifying individuals through small groups or combinations of attributes.

- Do not label a group negatively based on incomplete or biased data.
- Avoid charts that exaggerate differences through misleading axes or scales.
- Disclose exclusions and uncertainty that could change interpretation.
- Do not selectively present only results supporting a preferred conclusion.
- Use neutral language and distinguish measured behavior from assumptions about people.
- Consider who may be harmed if a result is misunderstood or used outside its intended context.

A technically correct calculation can still be communicated irresponsibly. Ethical communication includes both analytical accuracy and awareness of how a result may be used.

16. A Practical Language Guide

```
When evidence is direct:
    "In the records analyzed, ..."
    "The median was ..."
    "The proportion among eligible records was ..."

When evidence suggests but does not establish:
    "This pattern is consistent with ..."
    "One possible explanation is ..."
    "The difference may reflect ..."

When evidence is limited:
    "The available data do not show ..."
    "We cannot determine from this dataset ..."
    "This comparison should be interpreted cautiously because ..."

When describing association:
    "X was associated with Y in this dataset."
    "The variables showed a positive association."

Avoid unless justified:
    "X caused Y."
    "This proves ..."
    "Everyone ..."
    "Always ..."
    "The best group is ..."
```

17. Communication Quality Checklist

- Can the reader identify the question and main message quickly?
- Does every important number have a definition, unit, and denominator?
- Are observational unit, period, and analytical population clear?
- Are observation, interpretation, hypothesis, and causation kept distinct?
- Are the strongest findings prioritized?
- Are uncertainty and data-quality limitations visible?
- Have reasonable sensitivity checks been considered?
- Do charts have honest scales, readable labels, and useful captions?
- Are recommendations linked to evidence and appropriate to scope?
- Could a reader mistake a descriptive result for a causal or population-wide claim?
- Is confidential or identifying information protected?
- Can you explain the finding in plain language without relying on the chart alone?

18. Summary

- An insight combines evidence, context, meaning, boundaries, and implications.
- Prioritize findings by relevance, evidence strength, magnitude, reliability, and actionability.
- Communicate numbers with units, denominators, sample sizes, and meaningful comparisons.
- Separate observation, interpretation, hypothesis, and causal claims.

- Explain uncertainty and distinguish variability from uncertainty about a conclusion.
- Use sensitivity checks to see whether conclusions depend on analytical choices.
- Communicate group, time, and data-quality patterns with their specific caveats.
- Answer questions honestly; say what additional evidence is needed.
- Responsible communication protects people and prevents unsupported decisions.

19. Assignment

Part A — Concepts

- What makes a result an insight rather than just a number?
- Explain the difference between a percentage increase and a percentage-point increase.
- Distinguish observation, interpretation, hypothesis, and causal claim.
- Explain why variability and uncertainty are not the same thing.
- What does a sensitivity check tell us, and what does it not prove?
- Why can aggregate group comparisons be misleading?

Part B — Rewrite for responsible communication Rewrite each statement so that it is clear, appropriately cautious, and informative:

- 'The new website caused sales to increase.'
- 'Group A is twice as good as Group B.'
- 'The average is 500, so a typical customer spends 500.'
- 'There were no orders on Sunday, so customers do not shop on Sundays.'
- 'The EDA proves that the campaign was successful.'

Part C — Explain uncertainty A dataset shows that customers using a mobile app have a higher average order count than customers not using the app. List at least four alternative explanations or limitations. Write a short paragraph explaining what can and cannot be concluded.

Part D — Communicate a finding Choose one finding from your earlier EDA work. Write it in five parts: observation, evidence, interpretation, caveat, and next step. Then rewrite it as a two-sentence explanation for a non-technical client.

20. Coding and Communication Project — Insight Brief

Objective: create a concise insight brief from the EDA report developed in Chapter 24.

- Select three findings that matter most to the original analytical question.
- For each finding, provide a quantitative measure, denominator, and supporting figure or table.
- Separate direct observation from interpretation and possible explanation.
- Identify at least one uncertainty or limitation for each major finding.
- Perform and describe at least two sensitivity or robustness checks where appropriate.
- Write one evidence-linked next step for each finding.
- Prepare a one-page executive brief for a non-technical reader.
- Prepare a presentation outline of approximately five to seven slides.
- Prepare five likely client questions and evidence-based responses.
- Check that all numbers and claims match the reproducible notebook and Chapter 24 report.

Deliverables: one-page insight brief, presentation outline or slides, question-and-answer notes, and references to the supporting report and reproducible analysis.

Master notebook instruction: keep extending your single cumulative EDA master notebook with final checks and evidence references. The communication brief should use saved, traceable outputs rather than manually retyped or independently recalculated numbers.

Evaluation criteria: prioritization, clarity, quantitative accuracy, appropriate uncertainty language, separation of evidence and explanation, ethical awareness, and usefulness of next steps.

21. Preview of the Capstone

Project 1 — End-to-End Exploratory Data Analysis

You will bring the course together in one complete EDA engagement: define a question, understand the domain and observational unit, audit the data, explore variables and relationships, investigate unusual observations, document a reproducible workflow, create a professional report, and communicate the most useful insights with appropriate uncertainty.

The capstone is not a competition to use the largest number of plots or statistical measures. It demonstrates that you can make sound analytical choices, explain them, preserve a traceable workflow, and deliver a useful result.

Your final work should include a reproducible notebook/project, selected tables and figures, a professional EDA report, and a concise insight brief. The dataset and question should determine the methods—not the other way around.